Representations of Finite Groups of Lie Type

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1 Representations of $\mathbf{GL}_n(\mathbb{F}_q)$

Let $R_n := R(\mathrm{GL}_n(\mathbb{F}_q))$ be the representaiton ring of $\mathrm{GL}_n(\mathbb{F}_q)$. It sits inside $C_n := C(\mathrm{GL}_n(\mathbb{F}_q))$ the space of $\mathbb{C}$-valued class functions on $\mathrm{GL}_n(\mathbb{F}_q)$. In fact $R_n \subset C_n$ is a lattice. We wish to study the ring R_n. In fact, we will study:

$$R = \bigoplus_{n\geq 0} R_n \subset \bigoplus_{n\geq 0} C_n = C$$

1.1 Operations on Groupoids

Let's quickly review how to push / pull function on a groupoid. Let $a : X \to Y$ be a morphism of groupoids. For a groupoid X, let $[X] = \pi_0 X$ be the isomorphism class of X.

*based on lectures by Zhiwei Yun

We define the **cardinality** of X to be:

$$|X| = \sum_{x \in [X]} \frac{1}{\mathrm{Aut}_X(x)}$$

This definition has the good property that for every action groupoid X/G, we have:

$$|X/G| = |X|/|G|$$

The function naturally pullback: $a^* f(x) = f(a(x))$. For pushforward, we need to consider the groupoid fibre (homotopy pullback):

$$\begin{array}{ccc} X_y & \longrightarrow & X \\ \downarrow & & \downarrow a \\ \{y\} & \longrightarrow & Y \end{array}$$

where $\{y\}$ is the trivial groupoid. Very concretely, the objects in X_y are pairs:

$$\{(x, \phi) \mid \phi : a(x) \simeq y\}$$

and morphisms are $(x, \phi) \xrightarrow{\alpha} (x', \phi')$ for $\alpha : x \xrightarrow{\sim} x'$ with the property that

$$\phi' \circ a(\phi) = \phi$$

Then we can define the pushforward of a function:

$$(a_! f)(y) = \sum_{(x,\phi) \in [X_y]} \frac{f(x)}{|\mathrm{Aut}_{X_y}(x, \phi)|}$$

This is treating each point in the groupoid X_y with a "density". As a sanity check:

1. For $a : \mathrm{pt} \to \mathrm{pt}/G$, the essential fibre is G and that:

   $$a_! c = |G| \cdot c$$

   for constant function c.

2. For $a : \mathrm{pt}/G \to \mathrm{pt}$, we have:

   $$(a_! c)(\mathrm{pt}) = \sum_{\phi \in \mathrm{pt}} \frac{c(\mathrm{pt})}{|\mathrm{Aut}_{\mathrm{pt}/G}(\mathrm{pt})|} = \frac{c}{|G|}$$

Moreover, we can compute:

$$\begin{aligned} (a_! f)(y) &= \sum_{x \in [X] : a(x) \simeq y} \frac{f(x)}{|\ker(\mathrm{Aut}_X(x) \to \mathrm{Aut}_Y(ax))|} \cdot \frac{\mathrm{Aut}_Y(a(x))}{|\mathrm{im}\ (\mathrm{Aut}_X(x) \to \mathrm{Aut}_Y(a(x)))|} \\ &= \sum_{x \in [X] : a(x) \simeq y} f(x) \frac{|\mathrm{Aut}_Y(y)|}{|\mathrm{Aut}_X(x)|} \end{aligned}$$

1.2 Hall Algebra

Let $\mathrm{GL}_n(\mathbb{F}_q)/\mathrm{Ad}$ be the set of conjugacy classes of GL_n. Then as a groupoid:

$$\mathrm{GL}_n(\mathbb{F}_q)/\mathrm{Ad} \simeq \{(V,T) \mid \dim_{\mathbb{F}_q} V = n, T : V \xrightarrow{\sim} V\}$$

But RHS can also be interpreted as the groupoid of length n-torsion sheaves on $\mathbb{G}_m/\mathbb{F}_q$. Therefore:

$$\mathrm{GL}_n(\mathbb{F}_q)/\mathrm{Ad} \simeq \mathrm{Tor}_n(\mathbb{G}_m)$$

as groupoids. Hence:

$$C_n \simeq \mathrm{Fun}(\mathrm{Tor}_n(\mathbb{G}_m))$$
$$C \simeq \mathrm{Fun}(\mathrm{Tor}_c(\mathbb{G}_m))$$

where $\mathrm{Tor}_c(\mathbb{G}_m)$ is the space of finite support torsion sheaves on $\mathbb{G}_m$. We will call C the **Hall algebra**.

Our first goal is to give a Hopf algebra structure on C. Let $n = n_1 + n_2$, consider the groupoid:

$$\mathcal{E}_{n_1,n_2}(\mathbb{G}_m) \coloneqq \{0 \to \mathcal{F}_1 \to \mathcal{F} \to \mathcal{F}_2 \to 0 \mid \ell(\mathcal{F}_i) = n_i\}$$

We have natural maps:

$$\begin{array}{ccc} & \mathcal{E}_{n_1,n_2}(\mathbb{G}_m) & \\ {\scriptstyle a}\swarrow & & \searrow{\scriptstyle b} \\ \mathrm{Tor}_n(\mathbb{G}_m) & & \mathrm{Tor}_{n_1}(\mathbb{G}_m) \times \mathrm{Tor}_{n_2}(\mathbb{G}_m) \end{array}$$

for a remebers $\mathcal{F}$ and b remembers $\mathcal{F}_1, \mathcal{F}_2$. Then we have:

$$m = a_! b^* : \mathbb{C}[\mathrm{Tor}_{n_1}] \otimes \mathbb{C}[\mathrm{Tor}_{n_2}] \to \mathbb{C}[\mathrm{Tor}_n]$$

Explicitly, this is given by:

$$(f_1 \cdot f_2)(\mathcal{F}) = \sum_{\mathcal{F}_1 \subset \mathcal{F}, \ell(\mathcal{F}_1) = n_1} f_1(\mathcal{F}_1) f_2(\mathcal{F}/\mathcal{F}_1)$$

Example. *Let us explicitly compute:*

$$a_! : \mathbb{C}[\mathcal{E}_{n_1,n_2}] \to \mathbb{C}[\mathrm{Tor}_n]$$

$$(a_! g)(\mathcal{F}) = \sum_{[0 \to \mathcal{F}_1 \to \mathcal{F} \to \mathcal{F}_2 \to 0]} g(\mathcal{F}_1, \mathcal{F}_2) \frac{|\mathrm{Aut}_Y(\mathcal{F})|}{|\mathrm{Aut}(\mathcal{F}_1 \to \mathcal{F} \to \mathcal{F}_2)|}$$

Note that an isomorphism class of the fibre is determined by $\mathcal{F}_1 \subset \mathcal{F}$, and $\mathcal{F}_1', \mathcal{F}_1$ give the same isomorphism class iff there exists $T : \mathcal{F} \to \mathcal{F}$ such that $T(\mathcal{F}_1) = \mathcal{F}_2$, therefore:

$$(a_! g)(\mathcal{F}) = \sum_{\mathcal{F}_1 \subset \mathcal{F}} g(\mathcal{F}_1, \mathcal{F}/\mathcal{F}_1) \frac{|\mathrm{Aut}(\mathcal{F})|}{\#\{\alpha \in \mathrm{Aut}(\mathcal{F}) \mid \alpha(\mathcal{F}_1) = \mathcal{F}_1\} \cdot \#\{\beta(\mathcal{F}_1) \in \mathrm{Gr}(n_1, \mathcal{F})\}}$$

which is nothing but: $\sum_{\mathcal{F}_1 \subset \mathcal{F}} g(\mathcal{F}_1, \mathcal{F}/\mathcal{F}_1)$ by orbit-stablizer.

Proposition 1.1. *The Hall algebra C is associative and commutative.*

Corollary 1.2. *If $\mathcal{F}$ can be written as an extension of $\mathcal{F}_2$ by $\mathcal{F}_1$, then it can be written as an extension of $\mathcal{F}_1$ by $\mathcal{F}_2$.*

On the other hand, we can define a coproduct:

$$m^* = b_! a^* : \mathbb{C}[\mathrm{Tor}_n(\mathbb{G}_m)] \to \bigoplus_{n_1+n_2=n} \mathbb{C}[\mathrm{Tor}_{n_1}(\mathbb{G}_m)] \otimes \mathbb{C}[\mathrm{Tor}_{n_2}(\mathbb{G}_m)]$$

Explicitly:

$$m^* f(\mathcal{F}_1, \mathcal{F}_2) = \sum_{0\to\mathcal{F}_1\to\mathcal{F}\to\mathcal{F}_2\to 0} \frac{1}{|\mathrm{Hom}(\mathcal{F}_2, \mathcal{F}_1)|} f(\mathcal{F})$$

Fact: m^* is an algebra morphism.

Now define a bilinear form on C by:

$$\langle f_1, f_2\rangle_{\mathrm{Tor}_n(\mathbb{G}_m)} = \sum_{\mathcal{F}\in[\mathrm{Tor}_n(\mathbb{G}_m)]} \frac{1}{|\mathrm{Aut}(\mathcal{F})|} f_1(\mathcal{F})\overline{f_2(\mathcal{F})}$$

This form is positive definite and we have:

$$\langle m(f_1 \otimes f_2), f_3\rangle_C = \langle f_1 \otimes f_2, m^* f_3\rangle_{C\otimes C}$$

This justifies the notation m^*.

Theorem 1.3. *C is a commutative and cocommutative Hopf algebra*

Proof. (of commutativity). Define the set of primitve elements of C to be:

$$\mathrm{Prim}(C) = \{x \in C \mid m^*(x) = x \otimes 1 + 1 \otimes x\}$$

Note that if $x \in C$ is of degree n, then:

$$m^*(x) = x \otimes 1 + \cdots + 1 \otimes x$$

Therefore if $x \in C_1$ then x is primitive. Now the key observation is the following. If $C_+ = \bigoplus_{n\geq 1} C_n$, then $\mathrm{Prim}(C) \perp (C_+)^2$. Indeed if $x \in \mathrm{Prim}(C)$, $y, z \in (C_+)^\perp$:

$$\langle x, yz\rangle = \langle m^* x, y \otimes z\rangle = \langle x \otimes 1 + 1 \otimes x, y \otimes z\rangle = \langle x, y\rangle\langle 1, z\rangle + \langle 1, y\rangle\langle x, z\rangle = 0$$

Now we will prove that $[x, y] = 0$ for all $x, y \in C$ by induction on $\deg x + \deg y$. Indeed it suffices to prove that $[x, y] \in \mathrm{Prim}(C)$ because then $[x, y] \in \mathrm{Prim}(C) \cap (C_+)^2 = 0$. Now:

$$m^*[x, y] = [m^* x, m^* y] = [x \otimes 1 + \sum u_i \otimes v_i + 1 \otimes x, y \otimes 1 + \sum s_j \otimes t_j + 1 \otimes y]$$

By induction hypothesis, we know:

$$m^*[x, y] = [x, y] \otimes 1 + 1 \otimes [x, y]$$

as required. □

Remark. *There is another proof of using Serre duality and the fact that:*

$$\mathcal{F}^\vee := \underline{\mathrm{Ext}}^1(\mathcal{F}, \omega_{\mathbb{G}_m}) \simeq \mathcal{F}$$

for any torsion sheaf $\mathcal{F}$.

1.3 Symmetric Functions

Let's consider the ring

$$\Lambda = \mathbb{Z}[x_1, x_2, \cdots]^{S_\infty} = \varprojlim \mathbb{Z}[x_1, \cdots, x_N]^{S_N}$$

and the transition map is given by sending $y_N \mapsto 0$. The grading on Λ comes from the grading on each $\mathbb{Z}[x_1, \cdots, x_N]$. In fact Λ has a Hopf algebra structure.

1. The multiplication is given by multiplying polynomials.

2. The comultiplication is given by:

$$m^* : \mathbb{Z}[x_1, x_2, \cdots]^{S_\infty} \to \mathbb{Z}[y_1, y_2, \cdots]^{S_\infty} \otimes \mathbb{Z}[z_1, z_2, \cdots]^{S_\infty}$$

$$f(x_1, x_2, \cdots) \mapsto f(y_1, z_1, y_2, z_2, \cdots)$$

It is clear that Λ is commutative and co-commutative.

A very important way to think of the the Hopf algebra Λ is through the **Frobenius characteristic map**:

$$\mathrm{ch}_n : R(S_n) \xrightarrow{\sim} \Lambda_n$$

$$V \mapsto \frac{1}{n!} \sum_{\sigma \in S_n} \chi_V(\sigma) p_{\mu(\sigma)}$$

where $\mu(\sigma)$ is the cycle type of σ and $p_\mu = p_{\mu_1} \cdots p_{\mu_k}$ for $p_m = x_1^m + x_2^m + \cdots$. Finally the we set:

$$\mathrm{ch} = \oplus \mathrm{ch}_n : \oplus R(S_n) \xrightarrow{\sim} \Lambda$$

This is actually an isomorphism of Hopf algebras, where the multiplication and comultiplication on LHS are given by induction and restriction.

Let's have a digression on the study of the Hopf algebra Λ. Let

$$e_i = \sum_{a_1 < a_2 < \cdots < a_i} x_{a_1} \cdots x_{a_i}$$

Then $\Lambda = \mathbb{Z}[e_1, e_2, \cdots]$ (free generators). One can check that:

$$m^*(e_n) = 1 \otimes e_n + e_1 \otimes e_{n-1} + \cdots + e_{n-1} \otimes e_1 + e_n \otimes 1$$

It follows that $m^* E = E \otimes E$ for $E = \sum_{n \geq 0} e_n \in \hat{\Lambda}$ (i.e. E is a group-like element).

Definition 1.4. A λ-ring is a commutative ring R together with a sequence of endomorphisms $\lambda^n : R \to R$ for $n \geq 0$ satisfying:

1. $\lambda^0(x) = 1$;

2. $\lambda(x) = x$;

3. $\lambda^n(x + y) = \sum_{i+j=n} \lambda^i(x) \lambda^j(y)$

Remark. *The third identity is precisely the identity for exterior power to satisfy. In particular, if we identify $\Lambda = K(\mathrm{pt}/\mathrm{GL}_\infty)$, then λ^n is the usual exterior power. Under our basis, we have $\lambda^n(e_1) = e_n$.*

Another basis for Λ is the following:

$$h_1 = \sum x_i, \quad h_2 = \sum_{i \leq j} x_i x_j, \cdots$$

and let $H = \sum_{n \geq 0} h_n \in \hat{\Lambda}$ then still we have

$$m^* H = H \otimes H$$

so H is also group like.

The third set of useful symmetric polynomials are power sums:

$$p_n = \sum_i x_i^n$$

Note that $\Lambda \neq \mathbb{Z}[p_1, p_2, \cdots]$, since there will be denominators for creating h_i or e_i. However it is true that:

$$\Lambda_{\mathbb{Q}} = \mathbb{Q}[p_1, p_2, \cdots]$$

But note that $m^*(p_n) = p_n \otimes 1 + 1 \otimes p_n$. In view of the λ-ring structure of Λ, p_n are hte Adams operations applied to $p_1 = e_1$.

An elegant way to relate these elemnts is through generating series. Let's set:

$$E(t) = \sum_{n \geq 0} e_n t^n = \prod (1 + x_i t)$$

$$H(t) = \sum_{n \geq 0} h_n t^n = \prod (1 + x_i t + x_i^2 t^2 + \cdots) = \prod \frac{1}{1 - x_i t}$$

$$P(t) = \sum_{n \geq 1} \frac{p_n}{n} t^n$$

It is clear that $H(t)E(-t) = 1$, and that:

$$-\log E(-t) = \log H(t) = \sum_i -\log(1 - x_i t) = \sum_i \sum_{k=1}^{\infty} \frac{(x_i t)^k}{k} = P(t)$$

Since $E(t)$ is group-like, and log of a group-like element is primitve, we deduce that $P(t)$ is primitve, hence each p_n is primitive.

1.4 Structure of the Representation Ring

Now we will define a product and coproduct structure on R to give it a Hopf algebra structure, such that tha embedding $R \hookrightarrow C$ is a Hopf algebra morphism.

1. The multiplication is given by parabolic induction. Let $n = n_1 + n_2$, and $P_{n_1,n_2} \subset \mathrm{GL}_n$ be the maximal parabolic of two blocks of sizes n_1, n_2, so that it has Levi $\mathrm{GL}_{n_1} \times \mathrm{GL}_{n_2}$. Now consider:
$$\mathrm{ind}_{P_{n_1,n_2}}^{\mathrm{GL}_n} : R_{n_1} \otimes R_{n_2} \to R_n$$
This is the multiplication.

2. Similar, we have the restriction map:
$$\mathrm{res}_{P_{n_1} \times P_{n_2}}^{\mathrm{GL}_n} : R_n \to R_{n_1} \otimes R_{n_2}$$
$$\pi \mapsto \pi^{U_{n_1,n_2}}$$
and the restriction is given by:
$$R_n \to \bigoplus_{n_1+n_2=n} R_{n_1} \otimes R_{n_2}$$
$$\pi \mapsto \sum_{n_1+n_2=n} \mathrm{res}_{P_{n_1,n_2}}^{\mathrm{GL}_n}(\pi)$$

More or less from definition, we know the Hopf algebra structures on C and R are the same. In particular:

Corollary 1.5. *R is commutative and cocommutative Hopf algebra over $\mathbb{Z}$.*

In particular, this implies:
$$\mathrm{ind}_{P_{n_1,n_2}(\mathbb{F}_q)}^{\mathrm{GL}_n(\mathbb{F}_q)}(\pi_1 \times \pi_2) = \mathrm{ind}_{P_{n_2,n_1}}^{\mathrm{GL}_n(\mathbb{F}_q)}(\pi_2 \times \pi_1)$$

Definition 1.6. We call an irreducible character $\pi \in R_n$ **cuspidal** if π is primitive. More concretely, this means that:
$$\mathrm{res}_{n_1,n_2}(\pi) = 0$$
for all choices of n_1, n_2. Let $\mathcal{C}$ be the set of all cuspdial representations of $\mathrm{GL}_n(\mathbb{F}_q)$, and this bijects with the irreducible elements in $\mathrm{Prim}(R)$.

It is reasonable to expect that $\mathcal{C}$ determines R as a Hopf algebra.

Definition 1.7. A **positive self-adjoint Hopf algebra** over $\mathbb{Z}$ is a graded Hopf algebra $H = \bigoplus_{n \geq 0} H_n$ with $H_0 = \mathbb{Z}$ and is equipped with a $\mathbb{Z}$-basis $\{\pi\}$ and a bilinear form $\langle -, - \rangle$ such that $\{\pi\}$ forms an orthogonal basis and m, m^* are adjoint under $\langle -, - \rangle$.

For a positive self-adjoint Hopf algebra H, let $\mathcal{C}(H)$ be the set of irreducible primitive elements. It is clear that $\mathcal{C} \subset R$ fits into this setting.

Theorem 1.8 (Zelevinski)**.** *Let H be a positive self-adjoint Hopf algebra over $\mathbb{Z}$.*

1. *If $\mathcal{C}(H) = \{x\}$ with $x \in H_d$ then $H \simeq \Lambda^{(d)}$, where:*

$$\Lambda = \mathbb{Z}[x_1, x_2, \cdots]^{S_\infty}$$

but with all degrees multiplied by d. Here $x = \sum x_i$

2. *In general, we have decomposition:*

$$H = \otimes'_{\rho \in \mathcal{C}(H)} H_\rho$$

where each H_ρ decomposes as $H_\rho = \oplus_{n \geq 0} H_{\rho,n}$ for:

$$H_{\rho,n} = \mathrm{Span}\{\pi \mid \pi \text{ appears in } \rho^n\}$$

where H_ρ is a positive self-adjoint Hopf algebra with $\mathcal{C}(H_\rho) = \{\rho\}$.

In the context of representation ring R, the ρ is the theorem is a cuspidal representation, and π appears in ρ^n exactly when $\pi \subset \mathrm{ind}(\rho \times \cdots \times \rho)$.

Now let $|\mathbb{G}_m| = \overline{\mathbb{F}_q}^\times/\mathrm{Fr}$ be the closed points of $\mathbb{G}_m$ (or more concretely, the set of irreducible monic polynomials $f(t) \in \mathbb{F}_q[t]$ such that $f(0) \neq 0$).

Definition 1.9. Let C_x be the Hall algebra of torsion sheaves supported at $\{x\}$.

We can write:

$$C = \otimes'_{x \in |\mathbb{G}_m|} C_x$$

and each $\mathcal{C}_x$ has a grading:

$$C_x = \oplus_{n \geq 0} C_{x,n}$$

where $C_{x,n}$ is the functions on torsion sheaves supported at x of length n. Note that C_x is a commutative and co-commutative Hopf algebra.

Now there is a Hopf algebra isomorphism:

$$\Lambda_{\mathbb{C}}^{(d_x)} \simeq C_x$$

$$h_n \mapsto \mathbf{1}_{\mathrm{Tor}(\mathbb{G}_m)_{x,n}}$$

Here $d_x = \deg(x)$ and $\Lambda_{\mathbb{C}}^{(d_x)}$ is a regrading of the algebra by multiplying all degrees by d_x; the right hand side is the characteristic function which sends any torsion sheaf of length n (w.r.t $k(x)$) supported at x to 1. Under this isomorphism:

$$e_n \mapsto q_x^{\binom{n}{2}} \mathbf{1}_{k(x)^{\oplus n}}$$

$$p_n \mapsto \sum_{\lambda \vdash n} \phi_{r(\lambda)-1}(q_x) \mathbf{1}_\lambda$$

where:

1. $\mathbf{1}_\lambda$ is the characteristic function of the torsion sheaf of Jordan type λ

2. $r(\lambda)$ is the number of parts in λ

3. $\phi_i(t) = (1-t)(1-t^2)\cdots(1-t^i)$

Now let's specialize Zelevinski's theorem in our setting. We take $H = R$ be our positive self-adjoint Hopf algebra, and $\mathcal{C} = \mathcal{C}(R)$ be the subsets of cuspdial representations.

Theorem 1.10. *For $\rho \in \mathcal{C}$, let's define $R(\rho)$ to be the $\mathbb{Z}$-span of irreducible characters appearing in ρ^n for some n.*

1. *Each $R(\rho)$ is a Hopf subalgebra of R, and $R = \otimes_{\rho\in\mathcal{C}} R(\rho)$ as Hopf algebras.*

2. *We have Hopf algebra isomorphisms:*

$$\Lambda^{(d_\rho)} \xrightarrow{\sim} R(\rho)$$

where d_ρ is the degree of ρ in R.

The second isomorphism involves two choices. We define the **Steinberg representation** $\mathrm{St}_n(\rho)$ to be the largest irreducible summand of ρ^n, and the **Speh representation** to be the smallest irreducible summand of ρ^n. And we will send:

$$e_n \mapsto \mathsf{St}_n(\rho)$$

$$h_n \mapsto \mathrm{Sp}_n(\rho)$$

Example. *Let $\rho = \mathsf{triv}_1 \in R_1$, then $\mathrm{Sp}_n(\mathsf{triv}_1) = \mathsf{triv}_n$, and $\mathsf{St}_n(\mathsf{triv}_1)$ is the usual Steinberg:*

$$\{f : \mathcal{F}\ell_n(\mathbb{F}_q) \to \mathbb{C} \mid \int_{\mathbb{P}^1(\mathbb{F}_q)} f = 0, \text{ for all simple } \mathbb{P}^1 \text{ in } \mathcal{F}\ell_n\}$$

Here a simple $\mathbb{P}^1$ is one obtained by fixing all but one remember of the flag. This has dimension $q^{\binom{n}{2}}$.

Let's briefly discuss the proof of Zelevinski's theorem. Recall we have a nice pairing on R. Now if $\rho \in \mathcal{C}$ is irreducible and primitive, some computation shows:

$$\langle \rho^2, \rho^2 \rangle = 2$$

It follows that ρ^2 decomposes into a direct sum of two irreducibles, and call these Speh and Steinberg.

Note that we have a natural map:

$$\mathrm{supp} \ : \mathrm{Tor}(\mathbb{G}_m) \to \mathrm{Div}^+(\mathbb{G}_m)$$

On the R side, each irreducible representation of some $\mathrm{GL}_n(\mathbb{F}_q)$ gives us a "divisor" on $\mathcal{C}$: a finite $\mathbb{Z}_{\geq 0}$-combination of elements in $\mathcal{C}$. This is called the **cuspidal support**. Concretely, for π an irreducible representation of $\mathrm{GL}_n(\mathbb{F}_q)$, there exists $P \subset \mathrm{GL}_n$ and a cuspidal representation σ of Levi L_P such that $\pi \subset \mathrm{ind}_P^{\mathrm{GL}_n}(\sigma)$. Note that L_P is a product of small GL_{n_i}'s, and σ is a tensor product of cuspidal σ_i of each GL_{n_i}. We thus define the cuspidal support to be those σ_i counting with multiplicities. Although this definition appears to depend on P, the pair (L, σ) is well-defined up to conjugation. Therefore we obtain a map:

$$\mathrm{Irr}(\mathrm{GL}(\mathbb{F}_q)) = \coprod_{n\geq 0} \mathrm{Irr}(\mathrm{GL}_n(\mathbb{F}_q)) \to \mathrm{Div}^+(\mathcal{C})$$

1.5 Cuspidal Data

Consider the Pontryagin dual

$$\widehat{\mathbb{F}_{q^n}^\times} = \mathrm{Hom}(\mathbb{F}_q^\times, \mathbb{C}^\times)$$

We wish to define an analogue of $\overline{\mathbb{F}_q^\times} = \cup_n \mathbb{F}_{q^n}^\times$. Note that for $n \mid m$, we have the norm map:

$$\mathrm{Nm} : \mathbb{F}_{q^m}^\times \to \mathbb{F}_{q^n}^\times$$

which induces:

$$\mathrm{Nm}^* : \widehat{\mathbb{F}_{q^n}^\times} \to \widehat{\mathbb{F}_{q^m}^\times}$$

Let's denote:

$$\mathbb{G}_m^*(\mathbb{F}_{q^n}) \coloneqq \widehat{\mathbb{F}_{q^n}^\times}$$

and $\mathbb{G}_m^*(\overline{\mathbb{F}_q}) = \varinjlim_{\mathrm{Nm}^*} \mathbb{G}_m^*(\mathbb{F}_{q^n})$. Now $\mathbb{G}_m^*(\overline{\mathbb{F}_q})$ carries a Frobenius action, so that:

$$\mathbb{G}_m^*(\overline{\mathbb{F}_q})^{\mathrm{Fr}^n} = \mathbb{G}_m^*(\mathbb{F}_{q^n})$$

And let's define:

$$|\mathbb{G}_m^*| = \mathbb{G}_m^*(\overline{\mathbb{F}_q})/\mathrm{Fr}$$

Theorem 1.11 (Green)**.** *There is a natural bijection:*

$$\mathcal{C} \leftrightarrow |\mathbb{G}_m^*|$$

On the right hand side, take $\xi \in |\mathbb{G}_m^*|$ and take $d = d_\xi$ minimal such that $\xi \in \widehat{\mathbb{F}_{q^d}^\times}$. This is called the degree of ξ. The above bijection sends $\mathcal{C}_d$ (cuspidal representations of $\mathrm{GL}_d(\mathbb{F}_q)$) bijectively onto elements of $|\mathbb{G}_m^*|$ of degree d.

Corollary 1.12. *The number of cuspidal representations of degree d equals the number of closed points of $\mathbb{G}_m^*$ of degree d, which is the same as the number of closed points of $\mathbb{G}_m$ of degree d.*

There is still more we want to know; for example, the transition matrix between the natural bases in C and the R. It is a general fact that:

$$C \simeq \mathrm{Sym}(\mathrm{Prim}(C)) \simeq \mathrm{Sym}(\mathrm{Prim}(R)_{\mathbb{C}})$$

Thus:

$$\mathrm{Prim}(C) \simeq \mathrm{Prim}(R)_{\mathbb{C}}$$

Now we have a decomposition:

$$\mathrm{Prim}(C) = \bigoplus_{x \in |\mathbb{G}_m|} \mathrm{Prim}(C_x)$$

Recall that:

$$\Phi_x : \Lambda_{\mathbb{C}}^{(d_x)} \simeq C_x$$

It follows that: $\mathrm{Prim}(C_x) = \langle p_1, p_2, \cdots \rangle$. Let us define:

$$\widetilde{p_n}(x) \coloneqq \begin{cases} \Phi_x(p_{n/d_x}), & \text{if } d_x \mid n; \\ 0, & \text{otherwise.} \end{cases}$$

This is a primitive element in C_n. Therefore:

$$\{\widetilde{p_n}(x) \mid x \in |\mathbb{G}_m|, d_x \mid n\}$$

forms a basis for $\mathrm{Prim}(C)$.

Let's now find a basis for $\mathrm{Prim}(R)$. Given any $\xi \in |\mathbb{G}_m^*|$, we can view it as an element of $\mathcal{C}$ by Green's theorem. Recall that $R(\xi)$ is the subalgebra spanned by irreducible compoments of ξ^n for all n, $R \simeq \otimes_\xi R(\xi)$, and we have an isomorphism:

$$\psi_\xi : \Lambda^{(d_\xi)} \xrightarrow{\sim} R(\xi)$$

We may define

$$\widetilde{p_n}(\xi) \coloneqq \begin{cases} \psi_\xi(p_{n/d_\xi}), & \text{if } d_\xi \mid n; \\ 0, & \text{otherwise.} \end{cases}$$

as before. And similarly:

$$\{\widetilde{p_n}(\xi) \mid \xi \in |\mathbb{G}_m^*|, d_\xi \mid n\}$$

forms a basis for $\mathrm{Prim}(R)_{\mathbb{C}}$.

Now the goal is to compute the change of basis matrix between $\widetilde{p_n}(x)$ and $\widetilde{p_n}(\xi)$.

Theorem 1.13 (Green, Springer-Zelevinski)**.** *Let $\xi \in |\mathbb{G}_m^*|$ and $d_\xi \mid n$. Then:*

$$\widetilde{p_n}(\xi) = (-1)^{n-\frac{n}{d_\xi}} \sum_{y \in \mathbb{F}_{q^n}^\times} \langle \widetilde{\xi}, y \rangle_n \widetilde{p_n}(y)$$

where $\widetilde{\xi} \in \mathbb{G}_m^*(\overline{\mathbb{F}_q})$ is any point in the Galois orbit ξ (so that $\widetilde{\xi} \in \mathbb{G}_m^*(\mathbb{F}_{q^{d_\xi}})$), and for $y \in \mathbb{F}_{q^n}^\times$:

$$\langle \widetilde{\xi}, y \rangle_n = \widetilde{\xi}(\mathrm{Nm}_{\mathbb{F}_{q^n}/\mathbb{F}_{q^{d_\xi}}}(y)) \in \mathbb{C}^\times$$

Up to sign, this is the **Fourier transform** between $\mathbb{F}_{q^n}$-points of $\mathbb{G}_m$ and $\mathbb{F}_{q^n}$-points of $\mathbb{G}_m^*$!

Example. *Let's take $\mathsf{St}_n \subset \mathrm{Ind}_{B_n}^{\mathrm{GL}_n}(\mathsf{triv})$. It is known that:*

1. *$\mathsf{St}_n(g) = 0$ unless g is semisimple.*
2. *$\mathsf{St}_n(1) = q^{\binom{n}{2}}$*

Then for $\sum_x m_x d_x = n$, we have:

$$\mathsf{St}_n(\bigoplus_{x \in |\mathbb{G}_m|} k(x)^{\oplus m_x}) = (-1)^{n - \sum m_x} \cdot \prod_{x \in |\mathbb{G}_m|} q_x^{\binom{m_x}{2}}$$

Remark. *There is still some things not totally understood.*

1. *We might try to study representations of other classical groups over $\mathbb{F}_q$. Then their R-analogues are actually modules over $R(\mathrm{GL})$. What is the structure of this module?*

2. *Now that we discussed the Fourier transform between $|\mathbb{G}_m|$ and $|\mathbb{G}_m^*|$, the natural next step is to replace $\mathbb{G}_m$ by an elliptic curve $E/\mathbb{F}_q$. On the other hand, we get part of the ellitpic Hall algebra. One can defined the R-side just as in Green, Springer-Zelevinski's theorem. What's the representation theoretic meaning of the R-side?*

Recall that the key step for classifying the representations of GL_n is to find the cuspidal ones. We saw the cuspidal representations of GL_n essentially corresponds to points in Pontryagin dual $\mathbb{G}_m^*$ with degree n that do not factor through any norm map to a smaller $\mathbb{F}_{q'}$, considered up to Galois action. From this data we can recover a cuspidal representation of $\mathrm{GL}_n(\mathbb{F}_q)$.

Let $\xi : \mathbb{F}_{q^n}^\times \to \mathbb{C}^\times$. In Green's theorem, we need to extend it to $\widetilde{\xi} : \overline{\mathbb{F}_q}^\times \to \mathbb{C}^\times$. Then we consider the class function $g \in \mathrm{GL}_n(\mathbb{F}_q) \mapsto \widetilde{\xi}(\lambda_1) + \cdots + \widetilde{\xi}(\lambda_n)$ for eigenvalues λ_i of g. This defines a class function, and a key observation of Green is that this function is a virtual character of $\mathrm{GL}_n(\mathbb{F}_q)$.

This is essentially Brauer lifting: we start with the vector representation $\mathrm{GL}_n(\mathbb{F}_q) \to \mathrm{Aut}(\mathbb{F}_q^{\oplus n})$, and we apply the character $\widetilde{\xi}$ to its character. Thus even though the representation may not lift to characteristic 0, we can still lift its character.

Proposition 1.14 (Brauer)**.** *Let G be a finite group with a class function $f : G \to \mathbb{C}$. Then f is a virtual character iff $f|_H$ are virtual characters for all $H \subset G$ taking the form of a cyclic group times an ℓ-group for ℓ prime.*

Lusztig was able to write down a virtual representation of $\mathrm{GL}_n(\mathbb{F}_q)$ whose character is the above Brauer lifting; in fact he gave a complex of representations, and exactly one term in this complex is the relevant cuspidal representation. Note that this does not use any algebraic geometry, but rather a finite field analogue of buildings.

2 Deligne-Lusztig Theory

Let $G/\mathbb{F}_q$ be a split, connected, reductive group. Deligne-Lusztig produced a variety X with an action $G(\mathbb{F}_q) \to \mathrm{Aut}(H_c^*(X))$.

2.1 Drinfeld's Curve

The starting point is Drinfeld's construction of cuspidal representations of $\mathrm{SL}_2(\mathbb{F}_q)$. Consider the following affine curve:

$$C = \{(x, y) \mid xy^q - x^q y = 1\} \subset \mathbb{A}^2_{\overline{\mathbb{F}_q}}$$

This is a degree $q+1$ plane curve with certain symmetries. One symmetry is scaling: we can multiply x, y by $t \in \mathbb{F}_{q^2}$ such that $t^{q+1} = 1$. Another symmetry is that $\mathrm{SL}_2(\mathbb{F}_q)$ acts linearly

on (x, y). It is evident that these two actions commute. Drinfeld then consider $H^1_c(C, \overline{\mathbb{Q}_\ell})$, which by the above carries commuting actions of $\mathrm{SL}_2(\mathbb{F}_q)$ and $(\mathbb{F}_{q^2}^\times)^{\mathrm{Nm}=1}$.

Let $\overline{C}$ be the compactification of C. We know $D = \overline{C} \setminus C \simeq \mathbb{P}^1(\mathbb{F}_q)$ are $q+1$ rational points. The excision sequence gives me:

$$0 \to H^0_c(C) \to H^0(\overline{C}) \to H^0(D) \to H^1_c(C) \to H^1(\overline{C}) \to 0$$

We know that $H^0_c(C) = 0$; $\dim H^0(\overline{C}) = 1$; $\dim H^0(D) = q+1$. Degree genus formula tells me $g_{\overline{C}} = \frac{q(q-1)}{2}$, and therefore $\dim H^1(\overline{C}) = q(q-1)$. Putting all these together we deduce $H^1_c(C)$ has dimension q^2.

Now given a character

$$\theta : (\mathbb{F}_{q^2}^\times)^{\mathrm{Nm}=1} \to \overline{\mathbb{Q}_\ell}$$

we therefore have an $\mathrm{SL}_2(\mathbb{F}_q)$-action on $H^1_c(C)_\theta$. It turns out that this is a cuspidal irreducible representation when $\theta^2 \neq 1$, and this construction will exhaust all cuspidal representations of $\mathrm{SL}_2(\mathbb{F}_q)$; if $\theta^2 = 1$, this will yield a reducible representation instead.

Now consider the natural map $\pi : C \to \mathbb{P}^1$ $(x, y) \mapsto [x : y]$. The fibres are exactly the orbits by the above scaling action. In other words, this is a $(\mathbb{F}_{q^2}^\times)^{\mathrm{Nm}=1}$-torsor over its image. This map is not surjective, and it misses exactly the rational points $\mathbb{P}^1(\mathbb{F}_q)$ as $[x : y] = [x^q : y^q]$ iff $x^q y - x y^q = 0$. Indeed, this extends to a cover $\overline{C} \to \mathbb{P}^1$ ramified over the missing points.

The $\mathbb{P}^1$ in the above construction should be interpreted as the flag variety of SL_2. We should also seek for a Lie-theoretic interpretation of Drinfeld's curve.

2.2 Degline-Lusztig Variety

Let $\mathbb{W}$ be the abstract Weyl group, and X be the flag variety. We denote F the q-Frobenius of X (we use that X is defined over $\mathbb{F}_q$).

Definition 2.1. Given $w \in \mathbb{W}$, we define

$$X(w) = \{B \in X \mid \text{rel. pos}(B, F(B)) = w\}$$

In other words, we have a Cartesian square:

$$\begin{array}{ccc} X(w) & \longrightarrow & \mathcal{O}(w) \\ \downarrow & & \downarrow \\ X & \xrightarrow{\mathrm{id} \times F} & X \times X \end{array}$$

In the functor of points language, for any $\mathbb{F}_q$-algebra R, we have the q-Frobenius map $X(R) \to X(R)$. This map is induced by functoriality from the map $\mathrm{Fr} : R \to R$. In the case of $G = \mathrm{SL}_2$, this recovers that map:

$$[x : y] \mapsto [x^q, y^q]$$

from Drinfeld's construction. Concretely:

$$X(s)(\overline{\mathbb{F}_q}) = \{\ell \subset \overline{\mathbb{F}_q}^2 \mid \ell \neq F(\ell)\} = \mathbb{P}^1(\overline{\mathbb{F}_q}) \setminus \mathbb{P}^1(\mathbb{F}_q)$$

Return to a general connected reductive G, let us observe various symmetries of $X(w)$. There is a conjugation a action of $G(\mathbb{F}_q)$ on $X(w)$: this is because for $g \in G(\mathbb{F}_q)$, $F(g) = g$. Therefore $G(\mathbb{F}_q)$ acts on the cohomology of $X(w)$, and we define the **Deligne-Lusztig Representation**:

$$R_w^1 \coloneqq \sum_{i\in\mathbb{Z}} (-1)^i H_c^i(X(w), \overline{\mathbb{Q}_\ell}) \in K_0(\mathrm{Rep}(G(\mathbb{F}_q), \overline{\mathbb{Q}_\ell}))$$

Example. *Note that $X(1) = X(\mathbb{F}_q)$ is discrete, and therefore:*

$$R_1^1 = H_c^0(X(\mathbb{F}_q), \overline{\mathbb{Q}_\ell}) = \mathrm{Fun}(X(\mathbb{F}_q), \overline{\mathbb{Q}_\ell}) = \mathrm{Ind}_{B(\mathbb{F}_q)}^{G(\mathbb{F}_q)}(\overline{\mathbb{Q}_\ell})$$

It's clear that the whole construction is insensitive to central isogeny. Let's now study the case for $G = \mathrm{SL}_n$ (or GL_n). Let's try to describe the variety $X(w)$ for $w = (12\cdots n)$.

Recall that X parametrizes full flags in an n-dimensional vector space V. We must understand how to find relative position of two flags.

The general recipe is the following. Given $V_\bullet, V'_\bullet$ two flags, we have two filtrations on V, which we can combine into a single filtration indexed by the partially ordered set $[n]^2$ by setting:

$$\mathrm{Fil}_{a,b}(V) = V_a + V'_b$$

This gives rise to graded pieces:

$$\mathrm{gr}_{a,b}(V) = \mathrm{Fil}_{a,b}(V)/\mathrm{Fil}_{a,b-1}(V) + \mathrm{Fil}_{a-1,b}(V)$$

So we have n^2 terms summing to n. In fact, each term is either 0 or 1. More precisely, for a fixed a, the sequence $\mathrm{gr}_{a,b}(V)$ is the associated graded for the induced filtration on the one-dimensional space V_a/V_{a-1}. Therefore for each a, there exists unique $b = w^{-1}(a)$ such that $\mathrm{gr}_{a,w^{-1}(a)}V \neq 0$. This defines a permutation w^{-1}, then w defines the relative position of these two flags.

Example. *Let $G = \mathrm{SL}_3$ and $w = (123)$, the relative position between the standard flag:*

$$0 \subset \langle e_1\rangle \subset \langle e_1, e_2\rangle \subset V$$

and w times the standard flag

$$0 \subset \langle e_2\rangle \subset \langle e_2, e_3\rangle \subset V$$

should be w. Indeed,

$$\mathrm{gr}_{1,3}, \mathrm{gr}_{2,1}, \mathrm{gr}_{3,2} \neq 0$$

So that $w^{-1} = (132) \implies w = (123)$

Specializing the above algorithm to the cyclic case, we observe that $V_\bullet$ and $V'_\bullet$ have relative position $w = (12\cdots n)$ means that:

1. $V_1 + V_1' = V_2$
2. $V_1 + V_2' = V_3$
3. $\cdots$

Now we can describe $X(w)$ explicitly. The above discussion implies that $X(w)$ contains flags of form:

$$(V_1 < V_1 + F(V_1) < V_1 + F(V_1) + F^2(V_1) < \cdots < V_1 + \cdots + F^{n-1}V_1 = V)$$

with the condition that:

$$V_1, F(V_1), \cdots F^{n-1}(V_1)$$

are linearly independent. Therefore $X(w) \hookrightarrow \mathbb{P}^{n-1}$ is an open subvariety. More precisely, the condition is equivalent to V_1 not being contained in any hyperplane defined over $\mathbb{F}_q$. Indeed, if V_1 were contained in some Frobenius invariant hyperplane H , then obviously $F^k V_1 \subset H$ for all k, so that they cannot be linearly independent. Conversely, if $V_1, \cdots, F^a V$ are linearly independent, we take $H = V_1 + FV_1 + \cdots + F^{a-1}V_1$. This is a Frobenius invariant subspace of V containing V_1.

In summary, we concludes that:

$$X(w) = \mathbb{P}^{n-1} \setminus \bigcup_{H \subset V} \mathbb{P}(H)$$

for $H \subset V$ goes over all hyperplanes defined over $\mathbb{F}_q$.

Now let's compute the cohomology of $X(w)$. Choose a coordinate $\mathbb{P}^{n-1} = \mathbb{A}^{n-1} \cup \mathbb{P}^{n-2}$ so that $\mathbb{P}^{n-2}$ is defined over $\mathbb{F}_q$ so we can discard it. Hence:

$$X(w) = \mathbb{A}^{n-1} \setminus \cup H$$

Another observation is that $X(w)(\mathbb{F}_q) = \emptyset$ by definition. Similar argument shows that $X(w)(\mathbb{F}_{q^r}) = \emptyset$ for $r \leq n-1$.

Now let's try to compute the cohomology of $X(w)$. Consider $j : X(w) \hookrightarrow \mathbb{A}^{n-1}$ to be the open inclusion, and $i : \cup H \hookrightarrow \mathbb{A}^{n-1}$ to be the complement. We thus have an exact sequence:

$$0 \to j_! \mathbb{Q}_\ell \to \mathbb{Q}_\ell \to i_* \mathbb{Q}_\ell \to 0$$

The last term can be understood using inclusion and exclusion. We have a resolution:

$$\cdots \to \bigoplus_{H_i, H_j} i_{H_i \cap H_j *} \mathbb{Q}_\ell^{\oplus ?} \to \bigoplus_{H} i_{H*} \mathbb{Q}_\ell \to i_* \mathbb{Q}_\ell \to 0$$

In general the i-th term takes form:

$$\bigoplus_{H_i \subset \mathbb{A}^{n-1}} i_{H_i, *} \mathbb{Q}_{\ell, H_i}^{\oplus ?}$$

for ? to be determined. Therefore there will be a spectral sequence:

$$E_2^{p,q} = H^p(H_c^q(X(w), \mathcal{F}^\bullet)) \Rightarrow H^{p+q}(X(w), i_* \mathbb{Q}_\ell)$$

where $\mathcal{F}^p$ is the p-th term in the resolution. But note that:

$$H_c^q(X(w),\mathcal{F}^p)=\begin{cases}\bigoplus_{H_p\subset\mathbb{A}^{n-1}} i_{H_p,*}\mathbb{Q}_{\ell,H_p}(n-1-p)^{\oplus ?}, & \text{if } q=2(n-1-p);\\ 0, & \text{otherwise.}\end{cases}$$

Hence by parity reason the spectral sequence degenerates, and:

$$H_c^{2(n-1)-i}(X(w))=\begin{cases}\mathbb{Q}_\ell(n-1-i)^{\oplus ?}, & \text{if } 0\le i\le n-1;\\ 0, & \text{otherwise.}\end{cases}$$

It turns out that for $H_i\subset\mathbb{A}^{n-1}$ a codimension i affine subspace, the term $\mathbb{Q}_{\ell,H_i}$ appears with multiplicity $q^{\binom{i}{2}}$. In particular, the last step is:

$$H_c^{n-1}(X(w))\simeq\bigoplus_{x\in\mathbb{A}^{n-1}(\mathbb{F}_q)}\mathbb{Q}_{\ell,x}^{q^{\binom{n}{2}}}$$

This space has dimension $q^{n-1}\cdot q^{\binom{n-1}{2}}=q^{\binom{n}{2}}$. It turns out that this space carry the Steinberg representation. But this description does not reveal the full $\mathrm{GL}_n(\mathbb{F}_q)$-action; instead, it only shows the action of affine transformations of $\mathbb{A}^{n-1}$, viewed as a subgroup of $\mathrm{PGL}_n(\mathbb{F}_q)$ - this is the mirabolic subgroup.

Remark. *It is actually reasonable not able to recover the full action of* $\mathrm{GL}_n(\mathbb{F}_q)$*: the Steinberg is not a permuation representation, and therefore there should not be such an explicit description.*

2.3 Deligne-Lusztig Stack

Our goal is to construct the analogy of Drinfeld's curve. More precisely, we seek for:

$$\widetilde{X}(w)\to X(w)$$

a torsor for $\mathbb{T}_w(k)$ where $\mathbb{T}_w$ is certain torus over k. Before the general construction, let's consider type A.

Example. *Let* $G=\mathrm{GL}_n$*,* $w\in S_n$*. Recall that:*

$$X(w)(\overline{k})=\{V_\bullet\in X(\overline{k})\mid V_\bullet\xrightarrow{w}F(V_\bullet)\}$$

The relative position w *condition equips us with isomorphism:*

$$\mathrm{gr}_{w(i)}V\simeq\mathrm{gr}'_i(V')=\mathrm{Fr}(\mathrm{gr}_iV)$$

Let $\widetilde{X}(w)$ *classify flags together with basis of* gr_i *such that the basis the basis is compatible with the above isomorphism. More precisely:*

$$\widetilde{X}(w)(\overline{k})=\{(V_\bullet,v_i\in\mathrm{gr}_iV_\bullet\setminus\{0\})\mid V_\bullet\xrightarrow{w}F(V_\bullet),v_{w(i)}=F(v_i)\}$$

Note that $\widetilde{X}(w)$ admits an action of:

$$\{(\lambda_1, \cdots, \lambda_n) \mid \lambda_{w(i)} = F(\lambda_i) = \lambda_i^q\} \subset \mathbb{G}_m^n(\overline{k})$$

Let's twist the Frobenius structure on $\mathbb{G}_m^n$. Let F_w acts on $\mathbb{G}_m^n(\overline{k})$ via:

$$(\lambda_1, \cdots, \lambda_n) \mapsto (\lambda_{w^{-1}(1)}^q, \cdots, \lambda_{w^{-1}(n)}^q)$$

Then the above sequence (λ_i) are precisely the F_w-fixed points of $\mathbb{G}_m^n(\overline{k}) = \mathbb{T}(\overline{k})$. Now let's use the Galois descent datum $(\mathbb{T}_{\overline{k}}, F_w)$ to define $\mathbb{T}_w$ a torus over k. Then $\widetilde{X}(w)$ has a right action by:

$$\mathbb{T}_w(k) = \mathbb{T}(\overline{k})^{F_w}$$

Note here $\mathbb{T}_w(k)$ actually acts on the scheme $\widetilde{X}(w)$ not only on its k-points (which may actually be zero as in the case of Drinfeld's curve). For $w = (12\cdots n)$ cyclic, the fixed points are (λ_i) such that:

$$\lambda_1^q = \lambda_2, \lambda_2^q = \lambda_3, \cdots, \lambda_n^q = \lambda_1$$

This requires $\lambda_1^{q^n} = \lambda_1 \iff \lambda_1 \in \mathbb{F}_{q^n}^\times$. So in this case:

$$\mathbb{T}_w = \mathbb{T}_{(12\cdots n)} = R_{\mathbb{F}_q^n/\mathbb{F}_q}(\mathbb{G}_m)$$

In general the torus $\mathbb{T}_w$ is product of Weil restrictions according to cycle type.

For general construction, we need to introduce Deligne-Lusztig stacks. The idea is that we wish to study the quotient stack:

$$\mathfrak{X} = [G(k) \setminus X]$$

By definition, for a k-scheme S we have:

$$\mathfrak{X}(S) = \{(\mathcal{E}_{G(k)}, \lambda) \mid \mathcal{E}_{G(k)}/S : G(k)\text{-torsor}, G(k)\text{-equivariant } \lambda : \mathcal{E}_{G(k)} \to X\}$$

Now we need to use the general machinery of *Frobenius descent*: a $G(k)$-torsor over S/k is the same data as a G-torsor $\mathcal{E}_G$ over S together with a G-equivariant isomorphism:

$$\phi : \mathcal{E}_G \to {}^\tau\mathcal{E}_G = F_S^*\mathcal{E}_G$$

where F_S is the absolute Frobenius on S. Under this language, λ becomes a G-equivariant morphism $\mathcal{E}_G \to X = G/B$. But what is such a map? Since X is a G-homogeneous space, we only need to choose a point on each fibre of $\mathcal{E}_G$, and two choices give the same map if they differ by a B-action. Therefore, a choice of such a map is the same data as a section σ for the associated flag bundle:

$$\mathcal{E}_G \times^G X \to S$$

Therefore:

$$\mathfrak{X}(S) = \{(\mathcal{E}_G, \sigma, \phi) \mid \mathcal{E}_G \in \mathrm{Bun}_G(S), \sigma : S \to \mathcal{E}_G \times^G X, \phi : \mathcal{E}_G \simeq {}^\tau\mathcal{E}_G\}$$

In fact, the data of the pair $\mathcal{E} = (\mathcal{E}_G, \sigma)$ is nothing but a *B-torsor* over S. This follows from the following general fact: if $H \subset G$ is a subgroup, then a H-bundle $\mathcal{E}_H/S$ together with a H-equivariant bundle map $\mathcal{E}_H \to \mathcal{E}_G$ is the same data as a section of $S \to \mathcal{E}_G \times^G G/H = \mathcal{E}_G/H$. In the case of GL_n, a B-bundle is just a bundle of complete flags.

So we can also write:

$$\mathfrak{X}(S) = \{(\mathcal{E}, \phi) \mid \mathcal{E} : B\text{-torsor on } S, \phi : \mathcal{E}_G \simeq {}^\tau\mathcal{E}_G\}$$

and similarly:

$$\mathfrak{X}_w(S) = \{(\mathcal{E}, \phi) \mid \phi(\mathcal{E}_G) \xrightarrow{w} {}^\tau\mathcal{E}_G\}$$

There is a natural map:

$$X(w) \to \mathfrak{X}_w$$
$$B \mapsto (\mathcal{E} = (G, B), \phi)$$

where G is the trivial torsor, and the Borel B is regarded as a section of the trivial bundle. The trivialization ϕ is simply the identity map between trivial G-bundles. In fact this map is a $G(k)$-torsor, i.e.

$$[X(w)/G(k)] \simeq \mathfrak{X}_w$$

The idea is that we wish to use this to define $\widetilde{X}(w)$ using this. Let $\mathbb{T} = B/U$ be the absolute Cartarn (this is *canonical*, independent of the choice of B). Then we have diagram:

$$\begin{array}{ccc} & \mathcal{E} & \\ \swarrow & & \searrow \\ \mathcal{E}_G & & \mathcal{E}_{\mathbb{T}} \end{array}$$

where $\mathcal{E}_{\mathbb{T}}$ can be interpreted as graded lines. If $(\mathcal{E}, \phi) \in \mathfrak{X}_w$, then the relative position w tells us "associated lines are same up to w-reindex". As a priori, we don't have a map: of $\mathbb{T}$-torsors:

$$\mathcal{E}_{\mathbb{T}} \to {}^\tau\mathcal{E}_{\mathbb{T}}$$

But they being in $\mathfrak{X}_w$ gives:

$$\begin{array}{ccc} \mathbb{T} & \xrightarrow[w^{-1}]{} & \mathbb{T} \\ \curvearrowright & & \curvearrowright \\ \mathcal{E}_{\mathbb{T}} & \xrightarrow{\sim} & {}^\tau\mathcal{E}_{\mathbb{T}} \end{array}$$

Definition 2.2. Let

$$\mathrm{FBun}_G(S) = \{(\mathcal{F}, \phi) \mid \mathcal{F} \text{ is } G\text{-torsor on } S, \phi : \mathcal{F} \xrightarrow{\sim} {}^\tau\mathcal{F}\}$$

$$\mathrm{FBun}_{\mathbb{T}}^w(S) = \{(Q, \phi) \mid Q \ \mathbb{T}\text{-torsor on } S, \phi : Q \xrightarrow{\sim} {}^\tau Q \ w^{-1}\text{-twisted isomorphism of } \mathbb{T}\text{-torsor}\}$$

Remark. *We have:*

$$\mathrm{FBun}_G = [\mathrm{pt}/G(k)]$$

This really says any $(\mathcal{F}, \phi) \in \mathrm{FBun}_G(S)$ is the pullback of some G-torsor over $\mathbb{F}_q$, with identity map, at least étale locally on S. Similarly we have:

$$\mathrm{FBun}_{\mathbb{T}}^w = [\mathrm{pt}/\mathbb{T}_w(k)]$$

where $\mathbb{T}_w(k) = (\mathbb{T}_w(\overline{k}))^{F_w = F \circ w^{-1}}$

Therefore we get a map:

$$\mathfrak{X}(w) \to \mathrm{FBun}_G \times \mathrm{FBun}_{\mathbb{T}}^w = [\mathrm{pt}/G(k) \times \mathbb{T}_w(k)]$$

$$(\mathcal{E}, \phi) \mapsto (\mathcal{E}_G, \mathcal{E}_{\mathbb{T}})$$

Definition 2.3. We define $\widetilde{X}(w)$ to be the pullback:

$$\begin{array}{ccc} \widetilde{X}(w) & \longrightarrow & \mathrm{pt} \\ \downarrow & & \downarrow \\ \mathfrak{X}(w) & \longrightarrow & [\mathrm{pt}/G(k) \times \mathbb{T}_w(k)] \end{array}$$

Therefore, we have the diagram:

$$\begin{array}{ccc} \widetilde{X}(w) & \longrightarrow & \mathrm{pt} \\ \downarrow{\scriptstyle \mathbb{T}_w(k)\text{-torsor}} & & \downarrow \\ X(w) & \longrightarrow & \mathrm{pt} \times \mathrm{FBun}_{\mathbb{T}}^w \\ \downarrow{\scriptstyle G(k)\text{-torsor}} & & \downarrow \\ \mathfrak{X}(w) & \longrightarrow & \mathrm{FBun}_G \times \mathrm{FBun}_{\mathbb{T}}^w \end{array}$$

2.4 Deligne-Lusztig Representations

Let $k = \mathbb{F}_q$. Let's list the desired properties of the virtual characters:

$$R_w^1 = \sum (-1)^i H_c^i(X(w))$$

1. R_w^1 only depends on the conjugacy class of $w \in \mathbb{W}$. However, $X(w)$ very much depends on w. In fact $\dim X(w) = \ell(w)$.

2. Orthogonality:
$$\langle R_w^1, R_{w'}^1 \rangle = \#\{u \in \mathbb{W} \mid uwu^{-1} = w'\}$$

Example. *Let $G = \mathrm{SL}_2$. Then $X(1) = \mathbb{P}^1(k)$, $X(s) = X \setminus \mathbb{P}^1(k)$. It follows that:*

$$R_1^1 = \mathrm{Ind}_{B(k)}^{G(k)}(\overline{\mathbb{Q}_\ell}) = \mathsf{triv} + \mathsf{St}$$

$$R_s^1 = -[H_c^1(X(s))] + [H_c^2(X(s))] = -\mathsf{St} + \mathsf{triv}$$

It is clear that:

$$\langle \mathsf{triv} + \mathsf{St}, -\mathsf{St} + \mathsf{triv} \rangle = 0$$

Definition 2.4. An irreducible representation ρ is **unipotent** if:

$$\langle \rho, R_w^1 \rangle \neq 0$$

Remark. *This is equivalent to ρ appearing in some $H_c^i(X(w))$. This requires to prove that the components in R_w^1 are disjoint, by using certain purity argument we will see.*

Example. *Let $G = \mathrm{GL}_n$, recall that we established:*

$$\mathrm{Irr}(\mathrm{GL}_n(k)) \to \mathrm{Div}_n^+(\mathbb{G}_m^*)$$

where we view $\mathbb{G}_m^*$ as a "fake" algerbaic curve. The irreducible representation ρ of $\mathrm{GL}_n(k)$ is unipotent iff $\mathrm{supp}(\rho) = u \cdot [1]$ for some $u \in U_n$. This is equivalent to saying that ρ appears in $\mathrm{Ind}_{B_n(k)}^{\mathrm{GL}_n(k)}(\overline{\mathbb{Q}_\ell}) \iff \langle \rho, R_1^1 \rangle \neq 0$.

In general, it is not true that all unipotent representations ρ appear in R_1^1. But this serves as a good intuition.

The way to get other characters is the following.

Definition 2.5. For character $\theta : \mathbb{T}_w(k) \to \overline{\mathbb{Q}_\ell}^\times$, we can define a virtual character:

$$R_w^\theta \coloneqq \sum_i (-1)^i H_c^i(\widetilde{X}(w))_\theta \in K_0(\mathrm{Rep}(G(k), \overline{\mathbb{Q}_\ell}))$$

Let's list desired properties of R_w^θ.

1. R_w^θ only depends on the conjugacy class of (w, θ). More precisely, if $uwu^{-1} = w'$ for some $u \in \mathbb{W}$, then this u induces:

   $$\mathbb{T}_w(k) \xrightarrow{\sim} \mathbb{T}_{w'}(k)$$

   given by $t \mapsto u \cdot t$. This changes:

   $$\theta : \mathbb{T}_{w'}(k) \to \overline{\mathbb{Q}_\ell}^\times$$

   to $\theta \circ u^{-1} : \mathbb{T}_w(k) \to \overline{\mathbb{Q}_\ell}$. So the conjugation action should be written as:

   $$u \cdot (w, \theta) = (uwu^{-1}, \theta \circ u^{-1})$$

2. Weak Orthogonality:

   $$\langle R_w^\theta, R_{w'}^{\theta'} \rangle = \#\{u \in \mathbb{W} \mid u \cdot (w, \theta) = (w', \theta')\}$$

   In particular, if w, w' are not $\mathbb{W}$-conjugate, then $R_w^\theta, R_{w'}'$ are orthogonal.

3. Strong orthogonality: Consider:

   $$\bigoplus_i H_c^i(\widetilde{X}(w))_\theta, \quad \bigoplus_j H_c^j(\widetilde{X}(w'))_\theta$$

   They will be orthogonal if θ, θ' are not geometrically conjugate.

4. For any irreducible representation ρ, we can find R_w^θ such that:

$$\langle \rho, R_w^\theta \rangle \neq 0$$

This is even stronger than its appearance in some H_c^i since cancellation can happen. Note this is only useful if R_w^θ is "almost irreducible".

5. R_w^θ is indeed "almost irreducible". More precisely, for θ not fixed by any $u \in Z_{\mathbb{W}}(w)$ (generic θ), we have

$$\langle R_w^\theta, R_w^\theta \rangle = 1 \implies R_w^\theta = \pm \text{ irred char}$$

The main tool for proof of these properties will be **Deligne-Lusztig Fixed Point Formula**. This allows us to compute:

$$\mathrm{Tr}(s, R_w^\theta)$$

for semisimple s.

Now the question is, based on these properties, how should we organize $\mathrm{Irr}(G(k))$? THe guiding example should be the case for $G = \mathrm{GL}_n$. Recall in that case, we defined a "fake variety" $\mathbb{G}_m^*$ such that $\mathbb{G}_m^*(\overline{\mathbb{F}_q})$ has a Fr-action.

For general G, the recipe is the following. Let's define the "fake variety":

$$\mathbb{T}^* \coloneqq X^*(\mathbb{T}) \otimes_{\mathbb{Z}} \mathbb{G}_m^*$$

Note that there is a natural Frobenius action on $\mathbb{T}^*$. We then define:

$$\mathbb{T}_w^*(k) = \mathbb{T}^*(\overline{k})^{F_w}$$

Fact: We have $\mathbb{T}_w^*(k) \simeq \mathrm{Hom}(\mathbb{T}_w(k), \overline{\mathbb{Q}_\ell}^\times = \{\theta\}$ is the Pontrygain dual of $\mathbb{T}_w(k)$.

Now we see to classify irreducible representations of $G(k)$ according to θ.

Definition 2.6. Let $\theta : \mathbb{T}_w(k) \to \overline{\mathbb{Q}_\ell}^\times$, $\theta' : \mathbb{T}_{w'}(k) \to \overline{\mathbb{Q}_\ell}$. We say θ, θ' are geometrically conjugate if, after we view:

1. $\theta \in \mathbb{T}_w^*(k) \subset \mathbb{T}^*(\overline{k})$
2. $\theta' \in \mathbb{T}_{w'}^*(k) \subset \mathbb{T}^*(\overline{k})$

we have θ, θ' are in the same $\mathbb{W}$-orbit of $\mathbb{T}^*(\overline{k})$.

Corollary 2.7. *We get a map:*

$$\mathrm{Irr}(G(k), \overline{\mathbb{Q}_\ell}) \twoheadrightarrow \{\text{geom. conj. classes of } \theta\} = (\mathbb{T}^*(\overline{k})/\mathbb{W})^{\mathrm{Fr}}$$

So we can write:

$$\mathrm{Irr}(G(k), \overline{\mathbb{Q}_\ell}) = \coprod_{[\theta]} \mathrm{Irr}(G(k), \overline{\mathbb{Q}_\ell})_{[\theta]}$$

for $[\theta]$ running through geometric conjugacy classes. For $\theta = 1$, this gives unipotent representations.

2.5 Maximal Tori

Let G/F be a connected reductive group. Let's consider the set $\mathcal{T}(F)$ of all maximal tori *defined over* F. There is a map:

$$\sigma : \mathcal{T}(F)/G(F) \to \{\mathbb{W}\text{-torsors over Spec } F\}$$

as groupoids.

Let $\mathcal{T}$ be the moduli space of maximal tori of G. By definition for each F-algebra R, $\mathcal{T}(R)$ is the set of maximal tori $T \subset G_R$ defined over R. In fact $\mathcal{T} \simeq G/N_G(T_0)$ as a variety by choosing a base point T_0. On the other hand, let's consider the moduli space of pairs $T \subset B$ denoted as $\mathcal{TB}$. We have two forgetting maps:

$$\begin{array}{ccc} & \mathcal{TB} & \\ {\scriptstyle\tau}\swarrow & & \searrow{\scriptstyle\beta} \\ \mathcal{T} & & \mathcal{B} \end{array}$$

The map τ is a $\mathbb{W}$-torsor. Now given $T \subset G$ a torus defined over F, we can form a $\mathbb{W}$-torsor over Spec F via:

$$\begin{array}{ccc} \sigma(T) & \longrightarrow & \mathcal{TB} \\ \downarrow & & \downarrow \\ \text{Spec } F & \xrightarrow{T} & \mathcal{T} \end{array}$$

Lemma 2.8. *(Lang's theorem) Let $F = \mathbb{F}_q$, then:*

$$H^i(F, T) = \{1\}, \quad H^1(F, G) = \{1\}$$

for $i = 1, 2$

Lemma 2.9. *For $F = \mathbb{F}_q$, then σ is an isomorphism.*

Proof. Recall that G is split, so that the maximal tori of G defined over $\mathbb{F}_q$ up to conjugacy are in 1-to-1 correspondence with conjugacy classes of $\mathbb{W}$. The result then follows because:

$$H^1(F, \mathbb{W}) \simeq \mathbb{W}/\text{conj}$$

More concretely, let's fix a maximal torus $T \subset G_{\mathbb{F}_q}$ defined over $\overline{\mathbb{F}_q}$. Identify $\mathbb{W}$ with the Weyl group of T and lift w to $\dot{w} \in N_G(T)(\overline{\mathbb{F}_q})$. Then use $\text{Ad}(\dot{w})\text{Fr}$ as the descent datum for $G_{\overline{\mathbb{F}_q}}$ to get a $\mathbb{F}_q$-form G' together with a maximal torus T' that is the descent of T. Since $H^1(\mathbb{F}_q, G) = 1$, we see that $G' \simeq G$ over $\mathbb{F}_q$. We may thus view T' as a maximal torus of G, and check that it corresponds to conjugacy class of w. □

The next question is to understand $|\mathcal{T}(\mathbb{F}_q)|$.

Proposition 2.10. $|\mathcal{T}(\mathbb{F}_q)| = q^{2\nu}$, *where* $2\nu = \dim G - \text{rk}G = \#\{\text{roots}\}$.

Example. *Let's do a sanity check. Note that $G(\mathbb{F}_q)$ has order roughly $q^{\dim G}$; the normalizer of a fixed maximal torus T looks like:*

$$W \ltimes T(\mathbb{F}_q)$$

which has order roughly $q^{\mathrm{rk}G}$. On the other hand, let's explicitly compute the order in the case of $\mathrm{SL}_2(\mathbb{F}_q)$. Note first:

$$|\mathrm{SL}_2(\mathbb{F}_q)| = q(q^2-1)$$

Now there are two type of maximal torus T defined over $\mathbb{F}_q$:

1. *The split one, $T(\mathbb{F}_q) \simeq \mathbb{F}_q^\times \implies |N_G(T)(\mathbb{F}_q)| = 2(q-1)$. Therefore:*

$$|\mathrm{SL}_2(\mathbb{F}_q)|/|N_G(T)(\mathbb{F}_q)| = \frac{q(q+1)}{2}$$

2. *The nonsplit one, $T(\mathbb{F}_q) \simeq (\mathbb{F}_{q^2}^\times)^{\mathrm{Nm}=1} \implies |N_G(T)(\mathbb{F}_q)| = 2(q+1)$. Therefore:*

$$|\mathrm{SL}_2(\mathbb{F}_q)|/|N_G(T)(\mathbb{F}_q)| = \frac{q(q-1)}{2}$$

They indeed add up to q^2.

Proof. Let's use Grothendieck-Lefschetz trace formula for $\mathcal{T}$. Note that:

$$H_c^*(\mathcal{T}) = H_c^*(\mathcal{TB}/\mathbb{W}) \simeq H_c^*(\mathcal{TB})^{\mathbb{W}}$$

But recall that $\beta : \mathcal{TB} \to \mathcal{B}$ is an affine bundle of relative dimension ν. Hence:

$$H_c^*(\mathcal{TB}) \simeq H^*(\mathcal{B})[-2\nu](-\nu)$$

Note this isomorphism is not W-equivariant. In fact it will be equivariant if we twist one side by sign. Thus:

$$H_c^*(\mathcal{TB})^{\mathbb{W}} \simeq H^{2\nu}(\mathcal{B})[-2\nu](-\nu)$$

Since only the top degree contains the sign representation. But then RHS equals:

$$\overline{\mathbb{Q}_\ell}[-4\nu](-2\nu)$$

whose Frobenius trace is $q^{2\nu}$. □

2.6 Deligne-Lusztig Character Formula

We wish to compute:

$$\mathrm{Tr}(g, R_w^\theta) = \sum_i (-1)^i \mathrm{Tr}(g, H_c^i(\widetilde{X}(w))_\theta)$$

As a toy model, let's consider the following scenario. Let X be a finite simplicial complex with simplicial automorphism s. Then:

$$\sum_i (-1)^i \mathrm{Tr}(s, H^i(X, \mathbb{Q})) = \chi(X^s)$$

Why this is true? After subdivision, we may assume s permutes the simplices and acts by identity if it stablizes some complex. Then cohomology of X can be computed as the cohomology of singular cochain complex $C^i(X) = \mathbb{Q}[X(i)]$, where $X(i)$ is the set of i-simplices of X. Therefore LHS is noting but:

$$\sum_i (-1)^i \mathrm{Tr}(s, C^i(X))$$

Now s acts by permuting $X(i)$ so that $\mathrm{Tr}(s, C^i(X)) = |X(i)^s|$ and the result follows.

The actual statement of Deligne-Lusztig Fixed Point Formula is the following.

Theorem 2.11 (DL Fixed Point Formula)**.** *Let X be a scheme of finite type over an algebraically closed field k of characteristic p. Let g be a finite order automorphism of X, and write $g = su$ for s p'-part and u p-part. Then:*

$$\mathrm{Tr}(g, H_c^*(X)) = \mathrm{Tr}(u, H_c^*(X^s))$$

Remark. *There is nothing general we can say about u an automorphism with order power of p. For example take $X = \mathbb{A}^1$ and $g = u : x \mapsto x + 1$ which is an automorphism of order p. Now u acts trivially on $H_c^*(X) = \overline{\mathbb{Q}_\ell}[-2](-1)$ since it extends to a connected group action ($\mathbb{G}_a$-action), so it has trace 1; on the other hand $X^u = \emptyset$.*

As an application, we see that:

$$\mathrm{Tr}(g, H_c^*(X(w))) = \mathrm{Tr}(u, H_c^*(X(w)^s))$$

where $g = su$ is the Jordan decomposition of g. When $\theta \neq 1$, we have an analogous statement:

$$\mathrm{Tr}((g,t), H_c^*(\widetilde{X}(w))) = \mathrm{Tr}(u, H_c^*(\widetilde{X}(w)^{(s,t)}))$$

for $(g,t) \in G(k) \times \mathbb{T}_w(k)$. On the other hand, we also know:

$$\mathrm{Tr}(g, H_c^*(\widetilde{X}(w))_\theta) = \frac{1}{|\mathbb{T}_w(k)|} \sum_{t \in \mathbb{T}_w(k)} \mathrm{Tr}((g,t), H_c^*(\widetilde{X}(w)))\theta(t^{-1})$$

This allows us to relate all terms of $\mathrm{Tr}(g, R_w^\theta)$ to the space $\widetilde{X}(w)^{(s,t)}$. For simplicitly let's only consider the case when $\theta = 1$, so we are only interested in $X(w)^s$.

To describe $X(w)^s$, we will relate it to the Deligne-Lusztig variety for $L_s = C_G(s)^\circ$. We will consider the action of L_s on $\mathcal{B}^s$. In general $\mathcal{B}^s$ will be disjoint union of flag varieties for L_s.

Example. *Consider $G = \mathrm{GL}(V)$, and $s \in \mathrm{GL}(V)$ be semisimple. Then we have eigenvalue decomposition $V_\lambda = \bigoplus_\lambda V(\lambda)$. Then $\mathcal{B}^s$ is the moduli of flags respecting this decomposition:*

$$\mathcal{B}^s = \{V_\bullet \mid V_\bullet = \bigoplus_\lambda V_\bullet(\lambda), V_\bullet(\lambda) \subset V(\lambda)\}$$

Therefore we have a natural map:

$$\mathcal{B}^s \to \prod_\lambda \mathcal{B}_{\mathrm{GL}(V(\lambda))}$$

this is clearly surjective but not injective: given a collection of flags in $\mathcal{B}_{\mathrm{GL}(V)}$, there are multiple ways to shuffle the order that we add their components when assembling them into a flag in $\mathcal{B}^s$. It is cleaer that there are only finitely many ways for shuffling. More precisely, we have a map:

$$\mathcal{B}^s \to (\mathbb{G}_m)^n_{[s]} \subset \mathbb{T}$$

sending a flag $V_\bullet$ to the collection of eigenvalues of $\mathrm{gr}_i V_\bullet$, where $(\mathbb{G}_m)^n_{[s]}$ is the $\mathbb{W}$-orbit of s: the diagonal elements with the same eigenvalues as s, but possibly in different order. Note that this is a finite set, and each such set gives a unique way to lift an element of $\prod_\lambda \mathcal{B}_{\mathrm{GL}(V(\lambda))}$ to $\mathcal{B}^s$. Therefore we have an isomorphism:

$$\mathcal{B}^s \simeq (\mathbb{G}_m)^n_{[s]} \times \prod_\lambda \mathcal{B}_{\mathrm{GL}(V(\lambda))}$$

Note in this example, the connected components of $\mathcal{B}^s$ are in bijection with the $\mathbb{W}$-orbits of $s \in \mathbb{T}$. However, this is only true if $C_G(s)$ is connected, which is not true in general.

Example. *Consider $G = \mathrm{PGL}_2$, $s = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$. Clearly $\mathcal{B} \simeq \mathbb{P}^1$ and $\mathcal{B}^s = \{0, \infty\}$. Now $L_s = \mathbb{G}_m$ is the diagonal torus, and $C_G(s) = N_G(\mathbb{G}_m)$ acts on $\mathcal{B}^s$ by permuting the two points. So $(\mathbb{G}_m)_{[s]}$ is a singleton and does not distinguish the connected components of $\mathcal{B}^s$.*

In fact, to actually compute the connected components of $\mathcal{B}^s$, let T be a maximal torus containing s, $W = N(T)/T$, and $W(L_s) = N_{L_s}(T)/T \subset W$. Then the connected components of $\mathcal{B}^s$ are in bijection with:

$$W(L_s) \setminus W$$

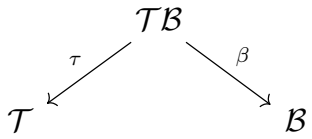
But our actual goal is to compute $X(w)^s$. Let's fix some maximal torus $T \subset G$ defined over $\mathbb{F}_q$, and let $B \in X(w)(\overline{k})$ with $T_{\overline{k}} \subset B$. Then for any $g \in L_s(k) \setminus G(k)$ with s with $s \in {}^gT$, with ${}^gT$ is a maximal torus in $L_s(k)$ and ${}^gB \cap L_s$ is a Borel subgroup of L_s. Let $w_g \in \mathbb{W}(L_s)$ be the relative position of ${}^gB \cap L_s$ and $F({}^gB \cap L_s)$.

Proposition 2.12. *Fix some maximal torus T as above. Then we have:*

$$X(w)^s = \coprod_{g \in L_s(k) \setminus G(k) : s \in {}^gT} X_{L_s}(w_g)$$

Proof. The key point is the proof of the proposition is the following statement: let $\pi \subset \mathcal{B}^s$ be a connected component such that $\pi \cap X(w) \neq \emptyset$, then π contains a borel $B \in X(w)$ over $\overline{k}$ such that B contains a maximal torus T defined over k. The reason for expecting this is that ${}^gT$ is a rational torus, so we would like to construct a rational torus inside $\pi \cap X(w)$ as well.

Let us prove just this key point. Recall that we have the diagram:



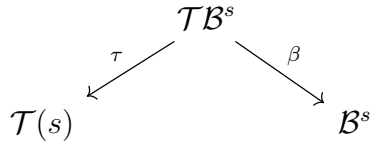
Recall that τ is a $\mathbb{W}$-torsor. Note that $\mathcal{TB}^s$ is the set $T \subset B$ such that s fixes both B and T. In particular, this implies $s \in B$ since s fixes B and $s \in N(T)$ since s fixes T. So

$$s \in N(T) \cap B = T$$

Therefore:

$$\mathcal{TB}^s = \{s \in T \subset B\}$$

This object fits into the diagram:



where $\mathcal{T}(s) = \{s \in T\} = \mathcal{T}_{L_s}$ (this is saying that a maximal torus inside L_s is nothing but a torus containing s). Now let $\widetilde{\pi} = \beta^{-1}(\pi) \subset \mathcal{TB}^s$ be the connected component of $\mathcal{TB}^s$ lying over π. Note we have two actions on $\pi_0(\mathcal{TB}^s)$: a left action by Frobenius and a right action of $\mathbb{W}$ coming from the structure of $\mathcal{TB}^s$ as a $\mathbb{W}$-torsor over $\mathcal{T}_{L_s}$.

By assumption, we have $\pi \cap X(w) \neq \emptyset$, so there is some $B \in \pi$ such that B, FB are in relative position w. Then we claim that $w^{-1} \circ F$ fixes $\widetilde{\pi}$. Indeed, for the above B, we have B, FB has relative postition w. Therefore we can find $T \in B \cap FB$ such that for $w \in N(T)/T$ we have $wB = FB$. Now consider $(T \subset B) \in \widetilde{\pi}$, then:

$$F \cdot (T \subset B) = (FT \subset FB), \quad w \cdot (T \subset B) = (T \subset wB) = (T \subset FB)$$

lies in the same component, because β has connected fibres. Therefore $w^{-1} \circ F$ defines another Frobenius structure on $\widetilde{\pi}$. Any Frobenius structure must have fixed points, so we conclude that there is some $(T \subset B) \in \widetilde{\pi}(\overline{k})$ such that $(FT \subset FB) = (T \subset B)w$. But this implies that $FT = T$ and hence T is defined over k, and B, FB are in relative position w, so $B \in X(w)$. □

From this we can deduce

Theorem 2.13 (DL Character Formula)**.** *Let T be as above, then:*

$$\mathrm{Tr}(su, R^{\theta}_{w}) = \sum_{g \in L_s(k)\backslash G(k), s \in {}^gT(k)} \theta(g^{-1}sg)\mathrm{Tr}(u, {}^{L_s}R^{1}_{w_g})$$

Here we identify $T(k)$ with $\mathbb{T}_w(k)$ via B, so that we can hit $g^{-1}sg \in T(k)$ by $\theta : \mathbb{T}_w(k) \to \overline{\mathbb{Q}_\ell}$. The main contribution:

$$^{L_s}Q_{w_g}(u) \coloneqq \mathrm{Tr}(u, {}^{L_g}R^1_{w_g})$$

is known as the *Green function*, which is the restriction of R^1_w to the unipotent elements. Note also that one can deduce from this theorem that R^θ_w depends only on the conjugacy class of (w, θ).

Lemma 2.14. *We have:*

$$\deg R^\theta_w = (-1)^{\ell(w)} \frac{|G(k)|_{(p')}}{|\mathbb{T}_w(k)|}$$

where $(-)_{(p')}$ denotes the p'-part of the number. In particular, this is independent of θ

Let's explain how to think of the lemma. Note that $|G(k)| = |G(k)/B(k)||B(k)|$, and that $|G(k)/B(k)| \equiv 1 (\mathrm{mod}\ q)$. It's also straightforward to check that $|B(k)| = (q-1)^r q^\nu$ for r the rank and ν the number of positive roots. Then we have:

$$\frac{|G(k)|_{(p')}}{|\mathbb{T}_w(k)|} = \frac{|G(k)|}{q^\nu \cdot |\mathbb{T}_w(k)|} \approx \frac{|G(k)|}{q^\nu q^r} \approx q^\nu$$

So all degrees of R^θ_w are similar to that of $R^1_1 = \mathrm{Ind}_{B(k)}^{G(k)}(\mathbb{C})$.

Proof. (sketched) Consider the Steinberg character:

$$\mathrm{Tr}(g, \mathsf{St}) = \begin{cases} \pm q^{\nu_{L_s}}, & \text{if } g = s \text{ semisimple;} \\ 0, & \text{otherwise.} \end{cases}$$

If $\theta \neq 1$, then by orthogonality:

$$0 = \langle \mathsf{St}, R^\theta_w \rangle = \sum_s (\cdots)$$

for some terms in the sum. But by the trace formula we know how to compute $\mathrm{Tr}(s, R^\theta_w)$, which gives us an equation for $\deg R^\theta_w$. □

2.7 Remaining Proofs

In the last section, we will prove:

1. The virtual representation R^θ_w only depends on the conjugacy class (this can be seen from the character formula, but let's supply a geometric proof).
2. The space R^θ_w exhaust all irreducible characters.

Let's first prove the first statement. The proof will base on a linear algebra trick showing two vectors are equal. Namely, given an inner product space $(V, \langle -, - \rangle)$ over $\mathbb{R}$ and $x, y \in V$, then:

$$x = y \iff \langle x, x \rangle = \langle x, y \rangle = \langle y, y \rangle$$

This is clear by considering $\langle x - y, x - y \rangle$.

Proposition 2.15. *If $(w,\theta),(w',\theta')$ are conjugate under the action of $\mathbb{W}$, then $R^\theta_w = R^{\theta'}_{w'}$*

Proof. We wish to use the trick above, however note that R^θ_w is $\overline{\mathbb{Q}_\ell}$-valued, not $\mathbb{R}$-valued. But here we actually work over a lattice, and we can just base change from $\mathbb{Z}$ to $\mathbb{R}$. Recall that we proved:

$$\langle R^\theta_w, R^\theta_w\rangle = \#\{u\in\mathbb{W} \mid u(w,\theta)=(w,\theta)\} := |\mathbb{W}(w,\theta)|$$

Similarly

$$\langle R^{\theta'}_{w'}, R^{\theta'}_{w'}\rangle = |\mathbb{W}(w',\theta')| = |\mathbb{W}(w',\theta')|$$

We also have:

$$\langle R^\theta_w, R^{\theta'}_{w'}\rangle = \#\mathrm{Transp}_{\mathbb{W}}((w,\theta),(w',\theta'))$$

This is a torsor for $\mathbb{W}(w,\theta)$, hence has cardinality equal to $|\mathbb{W}(w,\theta)|$. □

Remark. *If $\theta = 1$, then there is a direct proof not using orthogonality. Namely, consider w,w' in the same conjugacy class. Then there is some sequence of elementary moves to get from w to w'. It suffices to show these elementary moves do not change R^1_w.*

For example, consider the case when $w = sv, w' = vs$, and both are reduced expressions. Then given a flag $B\in X(w)$, we can find unique B' such that:

$$B \xrightarrow{s} B' \xrightarrow{v} FB$$

So we can interpret $X(w)$ as the set of such diagrams. Similarly,

$$X(w') = \{B \xrightarrow{v} B' \xrightarrow{s} FB\}$$

Nog given a point in $X(w)$, we can extend the diagram into:

$$B \xrightarrow{s} B' \xrightarrow{v} FB \xrightarrow{s} FB'$$

By forgetting B, we get a map $X(w)\to X(w')$. Similary we get a map $X(w')\to X(w)$. From the construction we see that the composition $X(w)\to X(w')\to X(w)$ is the Frobenius, and $X(w')\to X(w)\to X(w')$ is also the Frobenius. Thus we see that $X(w')$ is homeomorphic to $X(w)$ and thus they have the same cohomology.

There is a similar but more complicated proof to relatve $X(w), X(w')$ when w,w' differ by an elementary move but do not have the same length.

Now let's prove the exhaustion. The idea is to construct a linear combination of R^θ_w that is equal to δ_1. If that is the case, then suppose some irreducible character χ does not appear in $\{R^\theta_w\}$, we see that:

$$0 = \langle \chi, \delta_1\rangle = \frac{\chi(1)}{|G(k)|}$$

This is a contradiction. In fact, we will consider:

$$\frac{1}{|\mathbb{T}_w(k)|}\sum_\theta R^\theta_w$$

By Deligne Lusztig character formula, we know if $1 \neq s$ is semisimple:

$$\begin{aligned}\mathrm{Tr}(su, \sum_\theta R_w^\theta) &= \sum_\theta \sum_g \theta(g^{-1}sg)\mathrm{Tr}(u, {}^{L_s}R^1_{w_g}) \\ &= \sum_g \mathrm{Tr}(u, {}^{L_s}R^1_{w_g})(\sum_\theta \theta(g^{-1}sg)) = 0\end{aligned}$$

This means that $\frac{1}{\mathbb{T}_w(k)} \sum_\theta R_w^\theta$ is concentrated on unipotent elements, and:

$$\mathrm{Tr}(u, \sum_\theta R_w^\theta) = |\mathbb{T}_w(k)| \cdot \mathrm{Tr}(u, R_w^1)$$

because in this case $s = 1$ and $L_s = G(k)$ so we only sum over $g = 1$. In particular:

$$\frac{1}{|\mathbb{T}_w(k)|} \sum_\theta R_w^\theta = Q_w(u)$$

is the Green function. Therefore we will conclude the proof if we can prove:

Proposition 2.16. *We have the equality:*

$$\frac{1}{|\mathbb{W}|} \sum_w (-1)^{\ell(w)} Q_w = q^\nu \delta_1$$

Proof. We will use the same trick for proving two vectors are equal as before. Let L be the expression on LHS. We compute:

$$\langle L, L \rangle = \frac{1}{|\mathbb{W}|^2} \sum_{w,w'} (-1)^{\ell(w)+\ell(w')} \langle Q_w, Q_{w'} \rangle$$

Note that $\langle Q_w, Q_{w'} \rangle = 0$ unless $w \sim w'$, in which case $\ell(w) + \ell(w')$ is even. So:

$$\langle L, L \rangle = \langle \frac{1}{|\mathbb{W}|} \sum_w Q_w, \frac{1}{|\mathbb{W}|} \sum_w Q_w \rangle$$

Note that as a representation of $G(k)$, we have:

$$\sum_w R_w^1 = H^*(\mathcal{B})$$

and $G(k)$ acts trivially here (it extends to a G-action). So we get $\sum_w R_w^1 \simeq \mathsf{triv}^{\oplus |\mathbb{W}|}$. This implies that:

$$\frac{1}{|\mathbb{W}|} \sum_w Q_w = 1_{\mathcal{U}}$$

where $\mathcal{U}$ is the set of unipotent elements of $G(k)$. Therefore:

$$\langle L, L \rangle = \langle 1_{\mathcal{U}}, 1_{\mathcal{U}} \rangle = \frac{|\mathcal{U}|}{|G(k)|} = \frac{q^{2\nu}}{|G(k)|}$$

where we used the counting of unipotent elements in $G(\mathbb{F}_q)$. Next since we know the degree of R_w^θ, we compute that:

$$\langle L, q^\nu \delta_1 \rangle = \frac{q^\nu}{|G(k)||\mathbb{W}|} \sum_w (-1)^{\ell(w)} \deg R_w^1 = \frac{q^\nu}{|G(k)||\mathbb{W}|} \sum_w \frac{|G(k)|_{(p')}}{|\mathbb{T}_w(k)|} = \frac{1}{|\mathbb{W}|} \sum_w \frac{1}{|\mathbb{T}_w(k)|}$$

Now recall $\mathcal{T}$ is the moduli space of all maximal tori. Let's consider the groupoid $\mathcal{T}(k)/G(k)$. Then we know:

$$|\mathcal{T}(w)/G(k)| = \sum_{w \in \mathbb{W}/\text{conj}} \frac{1}{|N_{G(k)}(\mathbb{T}_w(k))|} = \sum_{w \in \mathbb{W}/\text{conj}} \frac{1}{|\mathbb{T}_w(k)||C_{\mathbb{W}}(w)|}$$

Therefore:

$$\frac{q^{2\nu}}{|G(k)|} = \frac{|\mathcal{T}(k)|}{|G(k)|} = \sum_{w \in \mathbb{W}} \frac{1}{|\mathbb{T}_w(k)||C_{\mathbb{W}}(w)|^{\mathbb{W}}w|} = \frac{1}{|\mathbb{W}|} \sum_{w \in \mathbb{W}} \frac{1}{|\mathbb{T}_w(k)|}$$

This shows $\langle L, q^\nu \delta_1 \rangle = \frac{q^{2\nu}}{|G(k)|}$. Moreover, $\langle q^\nu \delta_1, q^\nu \delta_1 \rangle = \frac{q^{2\nu}}{|G(k)|}$ is obvious. This finishes the proof. □